Game Development for Social Good

“Common Ground”

Project Proposal

1. Game title and developer

Title: Common Ground

Developer: Reinis Jānis Vaļūns (student ID: rv23040)

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2. Game concept

Common Ground is a small isometric pixel-art city builder in the browser. You are the mayor of Northhaven, a river city split into four districts: low-income housing, high-income housing, an industrial zone and a business district. You manage the budget, place and upgrade buildings, set taxes and policies, and react to events, in real time with pause and speed controls.

Each district has a faction with its own needs, and they often want opposite things. The core tension is money versus votes: the rich districts and businesses pay most of the taxes, but the low-income district has most of the voters. Spending in the wealthy areas grows your income faster; spending in the poor area wins more votes and reduces inequality, but pays back slowly.

At the start of each year you choose three challenges (your promises for the year). Each faction likes some challenges and dislikes others. In December the city votes, and you need more than half of the votes to stay mayor.

The city shows its condition on the map itself. Smog drifts over poor neighbourhoods, shops put up “FOR RENT” signs when business is bad, moving trucks appear when rents push people out, and streets fill up when people are doing well. The player reads the city by looking at it, not only through menus.

3. Story / theme

You have just been elected mayor of Northhaven, a fictional northern river city. The river splits it in two, and so does money. The north bank has the comfortable homes and the busy shopping streets. The south bank has the factories, the old coal power plant and the crowded housing next to them. One old bridge connects the two halves.

It is January, the city is divided, and you have one year before the next election.

Districts and factions

District	Faction	Residents (votes)	Tax income	Wants	Opposes
Millbrook (low-income housing)	Tenants' Union	50%	15%	Affordable rent, jobs, clean air, clinics, cheap transport	Rising rents, being ignored
Elmwood Heights (high-income housing)	Homeowners' Association	25%	25%	Low taxes, parks, quiet streets, rising property values	Social housing or industry nearby
Ironworks (industrial)	Industry Association	15%	25%	Cheap energy, roads, few regulations	Pollution limits and emission fees
Market Quarter (business)	Business Council	10%	35%	Customers, workers who can reach their jobs, fair business tax, deliveries	High business tax, closed and empty streets

The shares are starting values and will be tuned in playtesting. Ironworks' few residents are factory workers, so they care about jobs and clean air at the same time.

Rivals

Each rival courts particular factions, so neglecting a group gives a rival an opening:

- **The Builder** courts the Industry Association and Homeowners: low taxes, new roads, "jobs first".
- **The Idealist** courts the Tenants' Union: bold green and social promises that are often unfunded.
- **The Dealmaker** courts the Business Council and Homeowners: tax cuts and luxury riverside apartments.

Tone and style

Cosy isometric pixel art with an honest message. The city looks charming, but it shows its problems openly: haze, closed shops, protest signs. Rivals are fictional archetypes, not real politicians or parties.

4. Social good theme

- **The issue: inequality inside cities.** City decisions decide who gets clean air, affordable homes, good transport and jobs, and who carries the costs. The game focuses on how these problems connect:
- **Unequal pollution.** Industry and traffic often sit next to the poorest neighbourhoods, so the people who benefit least breathe the worst air.
- **Climate change.** The coal plant, the factories and car traffic release greenhouse gases. Cutting emissions costs money and industry support now, while the benefits arrive slowly and are shared by everyone.
- **Gentrification.** Improving a poor area can raise rents and push out the residents it was meant to help.
- **Access to jobs.** Without good public transport, people in poorer areas cannot reach work, and businesses cannot find workers.
- **A connected local economy.** When residents have less money, local shops lose customers and close, which costs more jobs.

- **Short-term thinking.** Tax income and votes arrive now, while the benefits of fairer and cleaner investments arrive later.

Related UN Sustainable Development Goals:

SDG	Target	How it appears in the game
10 Reduced Inequalities (primary)	10.2: social, economic and political inclusion of all	Money vs votes trade-off, inequality meter, factions comparing districts
11 Sustainable Cities and Communities (primary)	11.1 affordable housing, 11.2 public transport, 11.6 air quality, 11.7 green public spaces	Housing, rents, transit lines, bridge, parks, smog
13 Climate Action	13.2: climate measures in policies and planning; 13.3: awareness of how to cut emissions	Cutting city emissions: the yearly goal to cut factory smoke by 25%, factory filters, solar and wind replacing coal, buses replacing car trips
3 Good Health and Well-being	3.9: fewer illnesses from air pollution	Health of Millbrook residents under the smoke
7 Affordable and Clean Energy	7.2: more renewable energy	Replacing the coal plant with solar fields
8 Decent Work and Economic Growth	8.3: decent jobs and small businesses	Shop health in the Market Quarter, unemployment, retraining coal workers
16 Peace, Justice and Strong Institutions	16.7: responsive, inclusive, participatory decision-making	Yearly challenges, faction approval, elections

Why I chose this issue: about 55% of the world's population lived in cities in 2018, and the UN projects 68% by 2050 (UN DESA, 2018). Most people can see differences between richer and poorer neighbourhoods in their own city, but few think about how city decisions create or reduce them. City builders are a popular genre, but they usually treat residents as one happy-or-unhappy number. This game shows that "the city" is made of groups with different needs.

Who could benefit:

- **Young people and first-time voters**, who learn to ask "who benefits from this decision, and who pays?"
- **Teachers of civics, geography or economics**, who can use one 15 to 30 minute session as a classroom activity.
- **Local governments**, which benefit when citizens understand trade-offs and take part in decisions.

5. Social good integration into gameplay

The social issues are the game's rules, not its backdrop. Every social system had to pass three tests:

1. **Visible on the map:** the player can see the problem without opening a menu.
2. **A real trade-off:** fixing it costs money or upsets at least one faction.
3. **A delay:** the cost comes now, the benefit comes months later.

Mechanic	How it plays	What the player sees	Real-world idea	SDG
Money vs votes	Investing in Elmwood Heights or the Market Quarter raises tax income quickly. Investing in Millbrook wins more votes and narrows the gap, but pays back slowly.	Budget graph, approval per district	Who city spending serves	10.2
Fairness comparison	Factions judge you partly on what *other* districts get. Pouring everything into one district angers the rest. An	Inequality meter, protest signs at the town hall	Inequality is about relative, not only absolute, conditions	10.2

Mechanic	How it plays	What the player sees	Real-world idea	SDG
	inequality meter shows the quality-of-life gap between the best and worst district.			
Pollution drift and health	Smoke from Ironworks drifts over Millbrook. Health falls slowly, clinic costs rise and workers miss work. Fixes: factory filters (Industry Association angry), a green buffer park (slow), clean energy (expensive).	Haze over Millbrook, coughing residents	Pollution falls hardest on poorer areas	3.9, 11.6, 13
Shop health	Each shop needs customers (residents' spending power), workers who can reach it, and a fair business tax. Neglected shops struggle, then close; closed shops mean fewer jobs and less tax.	"SALE", then "FOR RENT" and "CLOSED" signs, dark windows, then reopening with scaffolding	The local economy depends on everyone having money to spend	8.3
Gentrification	Improving Millbrook raises land value and rents. Without a rent cap or social housing, original residents move away, and with them your voters.	Moving trucks, "NEW LUXURY FLATS" banners	Helping an area the wrong way can push out the people it was meant for	11.1
Transport to jobs	Most jobs are on the north bank, most workers on the south. Without bus, tram or a second bridge, commutes are long and unemployment rises. Car roads please the Business Council but add traffic and smog.	Traffic jams, crowded bus stops	Access to jobs depends on transport	11.2
Energy transition	The coal plant is cheap now and pollutes. Solar and wind cost more up front and save money later. Closing coal costs Ironworks jobs unless you fund retraining.	Plant smoke, new solar fields	Just transition to clean energy	7.2, 13
Yearly challenges and election	Challenges are promises with faction reactions. Keeping them raises approval; failing them gives rivals an opening.	Challenge cards, faction mood icons	Responsive, accountable local government	16.7

How it connects: the systems feed into each other. If the player ignores Millbrook to grow tax income, residents get sick and poorer, so they spend less, so Market Quarter shops close, so jobs and tax income fall. If the player helps Millbrook carelessly, rents rise and the residents are pushed out anyway. The winning approach is balanced, fair investment, which is exactly the game's message.

6. Intended impact / message

Core message: a city only works when it works for everyone. Decisions that favour the richest areas pay off quickly, but neglecting poorer areas eventually hurts the whole city.

After playing, players should:

- Understand that the costs of city decisions, like pollution, rent rises and long commutes, are not shared equally.
- Notice that helping a neighbourhood can backfire (gentrification) unless it protects the people who already live there.
- See that the local economy is connected: residents' incomes, shops, jobs and tax income depend on each other.
- Experience the mayor's dilemma: being popular and well-funded now versus fair and sustainable later.

- Ask of real decisions: “Who benefits, who pays, and who decides?”

How I will know whether the message comes through:

1. **Pre- and post-play questionnaire** (anonymous, about 5 minutes). The same 6 statements rated before and after playing, for example “Pollution affects all neighbourhoods equally” and “Improving a poor neighbourhood always helps its residents”. I expect answers to move toward the intended message.
2. **Open question after play:** “Describe one decision in the game where helping one group hurt another.” Answers show whether players noticed the trade-offs.
3. **In-game logging.** The game records how money was spent per district each month, the inequality meter at year end, and which challenges were chosen and kept. If players spread spending more evenly in later playthroughs, they are learning the trade-off.

7. Gameplay and rules

Mode: single-player, real-time with pause.

Time

- At normal speed, 1 month = 1 minute, so one year takes 12 minutes.
- Speed buttons: pause, 1x, 2x and 4x. At 4x a full year takes about 3 minutes, which makes it easy to demonstrate live.
- At the end of each month, taxes come in, upkeep is paid, and all indicators and faction moods update.
- The game pauses automatically for events and dilemmas so the player can decide calmly.

Structure of one year

1. **January:** choose 3 challenges from 6 to cards; the factions react right away.
2. **January to December:** govern for 12 months. Each month taxes come in, upkeep is paid and moods update; the player builds, sets taxes and policies, and answers events.
3. **December:** each district votes. With more than 50% of the votes the next year begins; otherwise the game is over.
4. **After year 4:** the City Report Card grades the city from A to F and shows the ending.

The demo will playsexactly one year.

Main player actions

Build on empty plots inside each district, from a building menu: social housing, apartments, park, clinic, school, bus stop, tram stop, second bridge, solar field, wind turbine, small shop, shopping mall.

Upgrade or demolish buildings, for example add filters to a factory or replace a car park with a park.

Set taxes per district: a residential tax for Millbrook and Elmwood Heights, a business tax for the Market Quarter and Ironworks.

Set policies: rent cap, emission limit, pedestrian zone, public transport fare.

Inspect any building or district to see its needs, mood and problems.

Respond to events, for example: a factory threatens to leave unless taxes drop; a developer wants to build luxury flats on Millbrook’s riverbank; the Business Council offers campaign money in exchange for a tax cut.

Yearly challenges

In January the game offers 5 to 6 challenges, and the player picks 3. Each card shows how every faction reacts:

Challenge	Tenants' Union	Homeowners	Industry	Business Council
Cut Ironworks smoke by 25%	++	+	−−	0
Build a tram line across the river	++	−	0	+
Build 50 social housing units	++	−	0	−
Lower business tax by 2%	−	+	+	++
Pedestrian zone in the Market Quarter	+	+	0	− at first, + later
Luxury flats on the riverside	−−	+	0	++

Choosing a challenge changes moods right away (people react to promises). Completing it by December gives a bigger boost; failing it makes the factions that wanted it angry and gives rivals an attack line.

Important rules and restrictions

- At most 3 challenges per year.
- Each building has a build cost, monthly upkeep and a build time (for example, a park takes 2 months and a tram line 5).
- Most effects are delayed: a clinic helps after 2 months, a school's effect on incomes appears only after many months.
- Each plot holds one building. Policies take effect from the next month.
- The coal plant can only close once clean energy covers the city's needs; closing it earlier causes blackouts.
- The budget may go into debt up to a limit. Three months in a row over the limit means bankruptcy.

How the election is decided

In December every district votes. Each faction's approval (0 to 100) comes from its own conditions, how fairly it is treated compared to the other districts, the year's challenges and events. Approval decides both how many residents support you over the rival and how many bother to vote.

You win with more than 50% of all votes cast. Because Millbrook has half of the residents, ignoring it is risky; but because it pays little tax, funding it takes money from somewhere else.

Typical flow of one year (the demo)

1. **January:** read the city report, which shows smog in Millbrook, two closed shops in the Market Quarter and a tight budget. Choose 3 challenges, for example the tram line, cutting smoke by 25%, and lower business tax.
2. **Spring:** start the tram line, raise the residential tax in Elmwood Heights slightly to pay for it, and watch the Homeowners' mood drop.
3. **Summer:** an event offers a shopping mall. Accepting brings tax income, but small shops nearby start showing "SALE" signs.
4. **Autumn:** the tram opens, unemployment in Millbrook falls, shops reopen, and the smog is lighter after factory filters.

5. **December:** election night shows the results district by district, then a short year report. Win, and the next year begins; lose, and the game ends.

8. Winning, losing and objectives

Main objective: stay mayor while making Northhaven fairer, cleaner and economically healthy.

Winning:

- **Demo and prototypes:** win the December election after one year.
- **Full game:** win 4 elections in a row (4 years).

After the last election the game shows a **City Report Card** with grades A to F for the inequality gap, air quality, housing affordability, employment, shop health and the budget. There are different endings:

- **"Common Ground":** you won and the city became fairer for everyone. The best ending.
- **"Divided City":** you won, but the gap between districts grew. You stayed in power, but the city paid for it.

Losing (game over):

- **Lose an election:** 50% of the votes or less.
- **Bankruptcy:** the budget stays over the debt limit for 3 months in a row.

On a loss the Report Card is still shown, so the player sees what they achieved and what went wrong.

Secondary objectives: the 3 yearly challenges, plus optional achievements such as "No closed shops for a whole year" or "Clean air in every district".

9. Platform and development tools

Platform: desktop web browser. The game runs from a link with nothing to install, so the instructor and playtesters can open it instantly. It is hosted for free on GitHub Pages.

Purpose	Tool
Game framework	Phaser 3, a free, open-source JavaScript framework for 2D browser games (MIT licence). It handles drawing, the camera, mouse input, animations and sound.
Language	JavaScript, with no build tools: one index.html plus a few script files
Map	Isometric ("2.5D") tile map with 2:1 tiles; buildings drawn back to front so they overlap correctly
Menus and panels	HTML and CSS on top of the game canvas: budget, faction moods, challenge cards, event pop-ups, election results
Art	Isometric pixel-art asset packs from itch.io or OpenGameArt (licence checked), adjusted in Piskel or LibreSprite; extra overlay sprites for signs, smog and moving trucks
Sound	Retro effects made with jsfxr, plus free chiptune music
Balancing data	All costs, effects and starting values in one JSON file, edited from a spreadsheet
Version control and hosting	Git with a public GitHub repository (as the course requires); GitHub Pages for the playable build
Playtest data	In-game logging exported as CSV; questionnaires in Google Forms
AI assistance	Claude (including Claude Code) to generate most of the code from this design document, plus debugging help; all use disclosed in the AI reflection

Development approach: the game is split into three parts that stay separate in the code:

- the simulation (money, factions, pollution, shops, elections) in plain JavaScript;
- the map drawn with Phaser;
- the menus in HTML.

I will build it in small steps, test each one in the browser, and commit to GitHub after every working step so I can roll back if AI-generated code breaks something. The game uses simple placeholder tiles first, with real pixel art added once the systems work.

10. Originality and existing games

Common Ground belongs to the city-builder genre, with ideas borrowed from political simulations. Its new contribution is putting **inequality between neighbourhoods** at the centre of the gameplay, through the money-versus-votes trade-off and consequences you can see on the map.

Game	What it does	How *Common Ground* differs
SimCity, Cities: Skylines	Large city builders focused on infrastructure, traffic and growth. Residents are mostly one mass of demand and happiness.	Much smaller, and built around four groups with conflicting needs. Success means fairness between districts, not just size and profit.
Tropico 4	You rule an island; factions have demands, and losing an election ends the game.	Similar faction and election idea, but in a democratic city, and each faction is tied to a place on the map with visible conditions.
Democracy series (Positech)	National policy simulator with sliders and overlapping voter groups; the player makes promises before elections.	Our challenges are similar to its promises, but *Common Ground* is a spatial city builder: you build on a map, and problems show up as smog, closed shops and moving trucks rather than numbers.
SimCityEDU: Pollution Challenge! (GlassLab, 2013)	Educational game where students play a mayor reducing pollution while keeping jobs and happiness up.	Shares the educational aim, but focuses on who suffers from pollution and adds housing, business and elections.

What makes this project different:

1. **Money versus votes.** The groups that pay the most and the groups that vote the most are different, so the player must think about who city spending serves.
2. **Unequal consequences.** Pollution, rent rises and long commutes fall on specific districts, not on "the city" as a whole.
3. **Help can backfire.** The gentrification mechanic shows that improving an area without protecting its residents pushes them out.
4. **A readable city.** Signs, haze and moving trucks let players spot problems by looking, which keeps the message from feeling like a lecture.
5. **Short sessions.** One year takes 12 minutes (3 minutes at 4× speed), so it fits a class or a quick demo.

11. Additional notes

Development plan

1. **Map:** isometric tiles with placeholder colours, the river, the 4 districts, camera movement and clicking on plots.
2. **Economy:** monthly clock, speed controls, budget, district taxes, placing buildings.

3. **Factions:** approval, needs and the fairness comparison, shown in HTML panels.
4. **Challenges and election:** challenge cards, one rival, election night results.
5. **Social systems:** pollution drift, shop health with signs, transport and jobs.
6. **Art, sound and polish:** pixel-art assets, overlays, tutorial, Report Card.
7. **Playtesting and balancing** using the logged data and questionnaires.

Steps 1 to 5 make up **Prototype II (due 10 November 2026)**, where the social good integration must be complete. Steps 6 and 7 lead to the **final version (due 1 December, presented 2 December 2026)**.

Main risks and how I handle them

- **AI-generated code breaks or becomes messy.** Small steps, testing after each, frequent GitHub commits, and keeping the simulation, map and menus separate.
- **Hard to balance.** All numbers live in one data file, so tuning does not require code changes.
- **The game feels preachy or political.** Players are never told what to do; they discover the trade-offs through consequences, and rivals are fictional archetypes, not real parties.

Use of AI

AI tools (mostly Claude) will be used to write code and to help with debugging and generating ideas for some of the events.

Sources

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